

João Rodrigues

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Education

Ph.D. in Computational Structural Biology <i>Bonvin Lab, Universiteit Utrecht, The Netherlands</i> Thesis: Computational Structural Biology of Macromolecular Interactions	2010 - 2014
M.Sc. in Biomolecular Sciences <i>Universiteit Utrecht, The Netherlands</i> Thesis: Predicting the Specificity of E2/E3 Complexes in the Ubiquitination Pathway	2008- 2010
B.Sc. in Biochemistry <i>Universidade de Coimbra, Portugal</i> Thesis: Concept-based Query Expansion for Gene Association Prediction from Literature Data	2004 - 2008

Research Experience

Schrödinger, NY, USA

Senior Principal Scientist	01.2026 - Present
Principal Scientist II • Scientific lead for several key products in the Schrödinger suite (e.g. Protein Preparation) • Led development of cryo-EM refinement and macrocycle modeling technologies. • Daily supervision of a team of 5 scientists and software engineers. • Co-authored 5 publications (3 as corresponding author).	06.2024 - 12.2025
Principal Scientist I	01.2023 - 06.2024
Senior Scientist II	01.2022 - 12.2022
Senior Scientist I • Led the development of a new method to find homologous ligand-bound structures. • Implemented novel algorithms improving the performance of flexible docking (IFD-MD). • Developed a method to model flexible linkers between protein domains. • Developed a protocol for flexible ligand docking for membrane protein targets. • Co-authored 2 publications (1 featured as cover of the journal Cell, vol 187, issue 3).	01.2021 - 12.2021

Stanford University, CA, USA

Postdoctoral Scholar, with Michael Levitt • Applied integrative modeling methods to study structure and pharmacology of GPCRs. • Co-authored 15 peer-reviewed publications (2 as corresponding author)	01.2016 - 12.2020
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Universiteit Utrecht, The Netherlands

Postdoctoral Scholar, with Alexandre Bonvin • Created and led the development of pdb-tools, a popular toolkit for structure manipulation. • Developed methods to predict the structure and affinity of protein-protein interactions. • Created materials for undergraduate and graduate courses on structural bioinformatics. • Co-authored 13 peer-reviewed publications.	04.2015 - 09.2015
Doctoral Candidate, with Alexandre Bonvin • Developed tools to predict the 3D structure of protein complexes from experimental data. • Contributed to the development of HADDOCK, a widely-used protein docking software. • Participated in several rounds of the CAPRI community challenge. • Co-authored 10 peer-reviewed publications.	10.2010 - 03.2015

Visiting Appointments

Institut de Biologie Physico-Chimique , Paris, France - Contributed to the development of VR-assisted methods for protein docking. - Acquired funding from the France-Stanford Center for Interdisciplinary Studies.	11.2017 - 12.2017
Department of Structural Biology, Weizmann Institute of Science , Israel - Helped solve the NMR structure of a peptide-mimic of chemokine CCL5 bound to CCR5. - Studied the role of post-translational tyrosine sulfation on chemokine recognition.	07.2017 - 08.2017
Department of Systems Biology, Harvard Medical School , MA, USA Analyzed large sequence datasets to derive spatial restraints for predicting the 3D structure of protein complexes by direct coupling analysis (co-evolution).	06.2012 - 07.2012
Department of Structural Biology, Stanford University , CA, USA Implemented a web server for protein structure refinement using statistical potentials.	10.2009 - 06.2010

Awards & Grants

The Molecular Sciences Software Institute - Seed Fellowship (\$25,000)	2020
ISMB/ECCB - RCSB PDB Poster Prize - Outstanding Student Paper	2019
Niels-Stensen Fellowship for Postdoctoral Studies (\$44,000)	2016
Stanford School of Medicine - Dean’s Postdoctoral Fellowship (\$25,000)	2016
Google Summer of Code Fellowship, Open Bioinformatics Foundation (\$5,000)	2010

Community Engagement

Invited Panelist , Sustainability Committee, Biophysical Society Meeting	2026
Invited Keynote , MICROBIOTEC’25, Azores (declined)	2025
Invited Panelist , Careers in Comp Chem, New York Area Group for Informatics and Modeling	2024
Co-Organizer , 3 rd EMBO Practical Course on “Integrative Modeling of Biomolecular Interactions”	2020
Guest , 2Scientists Podcast	2019
Scientific Committee , HIBIT Conference	2019-2025
Presenter , STEAM Festival Redwood City	2018-2019
Co-organizer , 2 nd EMBO Practical Course on “Integrative Modeling of Biomolecular Interactions”	2018
Speaker , Taste of Science Festival - San Francisco Bay Area	2017-2018
Co-organizer , 1 st EMBO Practical Course on “Integrative Modeling of Biomolecular Interactions”	2016
Presenter , NightLife, California Academy of Sciences	2016-2017
Volunteer , Bio-X Science Children’s Day - Stanford University	2016-2018
Co-organizer , 3 rd Meeting of Young Portuguese Researchers in Structural Biology (EJIBCE)	2015
Co-organizer , NMR Focus Workshop - “Methods for Protein Interactions and Drug Design”	2015
Invited Lecturer , Summer School “Exploring Nature’s Molecular Machines”, at Uni. Utrecht	2014
Invited Lecturer , “Molecular Machines”, B.Sc. Chemistry curriculum at Uni. Utrecht	2014
Invited Lecturer , M.Sc. Molecular and Cellular Life Sciences Introductory course at Uni. Utrecht	2014
Co-organizer , 1 st Meeting of Young Portuguese Researchers in Structural Biology (EJIBCE)	2013

Teaching

Instructor , Python and Machine Learning Virtual Classes	2024
Instructor , Software Carpentry	2019-Present
Teaching Assistant , HADDOCK User Workshops	2010 - 2021
Guest Lecturer , “Introduction to Bioinformatics”, B.Sc. Biology curriculum, Utrecht	2018-2019
Teaching Assistant , “Molecular Modeling and Math”, B.Sc. Chemistry curriculum, Utrecht	2018

Journal Service

Bioinformatics (ISSN: 1367-4811)	JCTC (ISSN: 1549-9618)
CSBJ (ISSN: 2001-0370)	J. Comput. Aided Mol. Des. (ISSN: 1573-4951)
Computers in Biology and Medicine (ISSN: 1879-0534)	Molecules (ISSN: 1420-3049)
F1000Research (ISSN: 2046-1402)	Nucleic Acids Research (ISSN: 0305-1048)
Genes (ISSN: 2073-4425)	PLoS Computational Biology (ISSN: 1553-7358)
IJMS (ISSN: 1661-6596)	PLoS One (ISSN: 1932-6203)
iScience (ISSN: 2589-0042)	Proteins (ISSN: 1097-0134)
Journal of Molecular Recognition (ISSN: 0952-3499)	SoftwareX (ISSN: 2352-7110)
JCIM (ISSN: 1549-960X)	The Journal of Physical Chemistry A (ISSN: 1520-5207)

Publications

indicates corresponding author, * indicates co-first author

46.

Zhang, L, Friesner, RA, Miller, EB, and **Rodrigues, JP**[#] (2026). *Generalization and Usability of Co-Folded GPCR–Ligand Complexes: A Physics-Guided Assessment*. Preprint

45.

Robson-Tull, J, and **Rodrigues, JP**[#] (2026). *Accurate Physics-Based Flexible Docking of Macrocyclic Ligands*. J. Med. Chem.

44.

Boyaci Selcuk, H, Reggiano, G, Robson-Tull, J, Zhang, L, and **Rodrigues, JP**[#] (2025). *Towards better structural models from cryo-electron microscopy data with physics-based methods*. FEBS Letters.

43.

Miller, EB, Hwang, H, Shelley, M, Placzek, A, **Rodrigues, JP**, et al. (2024). *Enabling structure-based drug discovery utilizing predicted models*. Cell.

42.

Honorato, RV, Trellet, ME, Jiménez-García, B, Schaarschmidt, JJ, Giulini, M, et al. (2024). *The HADDOCK2.4 web server for integrative modeling of biomolecular complexes*. Nat Protoc.

41.

Eastman, P, Galvelis, R, Peláez, RP, Abreu, CRA, Farr, SE, et al. (2024). *OpenMM 8: Molecular Dynamics Simulation with Machine Learning Potentials*. J. Phys. Chem. B.

40.

Coskun, D*, Lihan, M*, **Rodrigues, JP***, Vass, M, Robinson, D, et al. (2024). *Using AlphaFold and Experimental Structures for the Prediction of the Structure and Binding Affinities of GPCR Complexes via Induced Fit Docking and Free Energy Perturbation*. J. Chem. Theory Comput.

39.

Sorokina, M, Belapure, J, Tüting, C, Paschke, R, Papasotiriou, I, et al. (2022). *An Electrostatically-steered Conformational Selection Mechanism Promotes SARS-CoV-2 Spike Protein Variation*. Journal of Molecular Biology.

38.

Karakulak, T, Rifaioglu, AS, **Rodrigues, JP**, and Karaca, E (2021). *Predicting the Specificity- Determining Positions of Receptor Tyrosine Kinase Axl*. Front. Mol. Biosci.

37.

Jiménez-García, B, Teixeira, JMC, Trellet, M, **Rodrigues, JP**, and Bonvin, AMJJ (2021). *PDB-tools web: A user-friendly interface for the manipulation of PDB files*. Proteins.

36.

Fischhoff, IR, Castellanos, AA, **Rodrigues, JP**, Varsani, A, and Han, BA (2021). *Predicting the zoonotic capacity of mammals to transmit SARS-CoV-2*. Proceedings of the Royal Society B: Biological Sciences.

35.

Sorokina, M, M. C. Teixeira, J, Barrera-Vilarmau, S, Paschke, R, Papasotiriou, I, et al. (2020). *Structural models of human ACE2 variants with SARS-CoV-2 Spike protein for structure-based drug design*. Sci Data.

34.

Rodrigues, JP[#], Barrera-Vilarmau, S, M. C. Teixeira, J, Sorokina, M, Seckel, E, et al. (2020). *Insights on cross-species transmission of SARS-CoV-2 from structural modeling*. PLoS Comput Biol.

33.

Kaplan, M, Sweredoski, MJ, **Rodrigues, JP**, Tocheva, EI, Chang, Y-W, et al. (2020). *Bacterial flagellar motor PL-ring disassembly subcomplexes are widespread and ancient*. Proc. Natl. Acad. Sci. U.S.A.

32.

Roel-Touris, J, Don, CG, V. Honorato, R, **Rodrigues, JP**, and Bonvin, AMJJ (2019). *Less Is More: Coarse-Grained Integrative Modeling of Large Biomolecular Assemblies with HADDOCK*. J. Chem. Theory Comput.

31.

Rodrigues, JP[#], Teixeira, JMC, Trellet, M, and Bonvin, AMJJ (2018). *pdb-tools: a swiss army knife for molecular structures*. F1000Res.

30.

Pataki, CI, **Rodrigues, JP**, Zhang, L, Qian, J, Efron, B, et al. (2018). *Proteomic analysis of monolayer-integrated proteins on lipid droplets identifies amphipathic interfacial α -helical membrane anchors*. Proc. Natl. Acad. Sci. U.S.A.

29.

Masureel, M, Zou, Y, Picard, L-P, Van Der Westhuizen, E, Mahoney, JP, et al. (2018). *Structural insights into binding specificity, efficacy and bias of a β 2AR partial agonist*. Nat Chem Biol.

28.

Kurkcuoglu, Z, Koukos, PI, Citro, N, Trellet, ME, **Rodrigues, JP**, et al. (2018). Performance of HADDOCK and a simple contact-based protein–ligand binding affinity predictor in the D3R Grand Challenge 2. J Comput Aided Mol Des.

Publications (continued)

indicates corresponding author, * indicates co-first author

27. Bonvin, AMJJ, Karaca, E, Kastritis, PL, and **Rodrigues, JP** (2018). *Defining distance restraints in HADDOCK*. Nat Protoc.
26. Baro, B, Játiva, S, Calabria, I, Vinaixa, J, Bech-Serra, J-J, et al. (2018). *SILAC-based phosphoproteomics reveals new PP2A-Cdc55-regulated processes in budding yeast*. GigaScience.
25. Abriata, LA, **Rodrigues, JP**, Salathé, M, and Patiny, L (2018). *Augmenting Research, Education, and Outreach with Client-Side Web Programming*. Trends in Biotechnology.
24. Abayev, M, **Rodrigues, JP**[#], Srivastava, G, Arshava, B, Jaremko, Ł, et al. (2018). *The solution structure of monomeric CCL5 in complex with a doubly sulfated N-terminal segment of CCR5*. The FEBS Journal.
23. Visscher, KM, Medeiros-Silva, J, Mance, D, **Rodrigues, JP**, Daniëls, M, et al. (2017). *Supramolecular Organization and Functional Implications of K⁺ Channel Clusters in Membranes*. Angew Chem Int Ed.
22. Vangone, A, **Rodrigues, JP**, Xue, LC, Van Zundert, GCP, Geng, C, et al. (2017). *Sense and simplicity in HADDOCK scoring: Lessons from CASP-CAPRI round 1*. Proteins.
21. Komolov, KE, Du, Y, Duc, NM, Betz, RM, **Rodrigues, JP**, et al. (2017). *Structural and Functional Analysis of a β 2-Adrenergic Receptor Complex with GRK5*. Cell.
20. Karaca, E, **Rodrigues, JP**, Graziadei, A, Bonvin, AMJJ, and Carlomagno, T (2017). *M3: an integrative framework for structure determination of molecular machines*. Nat Methods.
19. Xue, LC, **Rodrigues, JP**, Kastritis, PL, Bonvin, AM, and Vangone, A (2016). *PRODIGY: a web server for predicting the binding affinity of protein–protein complexes*. Bioinformatics.
18. Xue, LC, **Rodrigues, JP**, Dobbs, D, Honavar, V, and Bonvin, AMJJ (2016). *Template-based protein–protein docking exploiting pairwise interfacial residue restraints*. Brief Bioinform.
17. Van Zundert, GCP, **Rodrigues, JP**^{*}, Trellet, M, Schmitz, C, Kastritis, PL, et al. (2016). *The HADDOCK2.2 Web Server: User-Friendly Integrative Modeling of Biomolecular Complexes*. Journal of Molecular Biology.
16. **Rodrigues, JP**, Melquiond, ASJ, and Bonvin, AMJJ (2016). *Molecular dynamics characterization of the conformational landscape of small peptides: A series of hands-on collaborative practical sessions for undergraduate students*. Biochem Molecular Bio Educ.
15. Liu, Y, **Rodrigues, JP**, Bonvin, AMJJ, Zaal, EA, et al. (2016). *New Insight into the Catalytic Mechanism of Bacterial MraY from Enzyme Kinetics and Docking Studies*. Journal of Biological Chemistry.
14. Lensink, MF, Velankar, S, Kryshafovych, A, Huang, S, Schneidman-Duhovny, D, et al. (2016). *Prediction of homoprotein and heteroprotein complexes by protein docking and template-based modeling: A CASP-CAPRI experiment*. Proteins.
13. Rad-Malekshahi, M, Visscher, KM, **Rodrigues, JP**, De Vries, R, Hennink, WE, et al. (2015). *The Supramolecular Organization of a Peptide-Based Nanocarrier at High Molecular Detail*. J. Am. Chem. Soc.
12. **Rodrigues, JP**, and Bonvin, AMJJ (2014). *Integrative computational modeling of protein interactions*. The FEBS Journal.
11. Kastritis, PL, **Rodrigues, JP**, Folkers, GE, Boelens, R, and Bonvin, AMJJ (2014). *Proteins Feel More Than They See: Fine-Tuning of Binding Affinity by Properties of the Non-Interacting Surface*. Journal of Molecular Biology.
10. Kastritis, PL, **Rodrigues, JP**, and Bonvin, AMJJ (2014). *HADDOCK2P2I : A Biophysical Model for Predicting the Binding Affinity of Protein–Protein Interaction Inhibitors*. J. Chem. Inf. Model.
9. Hopf, TA, Schärfe, CPI, **Rodrigues, JP**^{*}, Green, AG, Kohlbacher, O, et al. (2014). *Sequence co-evolution gives 3D contacts and structures of protein complexes*. eLife.
8. Ferguson, FM, Dias, DM, **Rodrigues, JP**, Wienk, H, Boelens, R, et al. (2014). *Binding Hotspots of BAZ2B Bromodomain: Histone Interaction Revealed by Solution NMR Driven Docking*. Biochemistry.
7. **Rodrigues, JP**, Melquiond, ASJ, Karaca, E, Trellet, M, Van Dijk, M, et al. (2013). *Defining the limits of homology modeling in information-driven protein docking*. Proteins.
6. Moretti, R, Fleishman, SJ, Agius, R, Torchala, M, Bates, PA, et al. (2013). *Community-wide evaluation of methods for predicting the effect of mutations on protein-protein interactions: Evaluation of Interaction Mutation Prediction*. Proteins.
5. Dias, DM, **Rodrigues, JP**, Domingues, NS, Bonvin, AMJJ, and Castro, MMCA (2013). *Unveiling the Interaction of Vanadium Compounds with Human Serum Albumin by Using 1 H STD NMR and Computational Docking Studies*. Eur J Inorg Chem.
4. **Rodrigues, JP**, Trellet, M, Schmitz, C, Kastritis, P, Karaca, E, et al. (2012). *Clustering biomolecular complexes by residue contacts similarity*. Proteins.
3. **Rodrigues, JP**, Levitt, M, and Chopra, G (2012). *KoBaMIN: a knowledge-based minimization web server for protein structure refinement*. Nucleic Acids Research.

Publications (continued)

indicates corresponding author, * indicates co-first author

2. De Vries, SJ, Melquiond, ASJ, Kastritis, PL, Karaca, E, Bordogna, A, et al. (2010). *Strengths and weaknesses of data-driven docking in critical assessment of prediction of interactions*. Proteins.
1. Matos, S, Arrais, JP, **Rodrigues, JP**, and Oliveira, JL (2010). *Concept-based query expansion for retrieving gene related publications from MEDLINE*. BMC Bioinformatics.

Book Chapters

2. Geng, C, Narasimhan, S, **Rodrigues, JP**, and Bonvin, AMJJ (2017). *Information-Driven, Ensemble Flexible Peptide Docking Using HADDOCK*. In Modeling Peptide-Protein Interactions: Methods and Protocols, O. Schueler-Furman and N. London, eds. (Springer).
1. **Rodrigues, JP**, Karaca, E, and Bonvin, AMJJ (2015). *Information-Driven Structural Modelling of Protein–Protein Interactions*. In Molecular Modeling of Proteins Methods in Molecular Biology., A. Kukol, ed. (Springer New York).

Links

Github	github.com/joaorodrigues
Google Scholar	scholar.google.com/citations?user=UeXRdzQAAAAJ
LinkedIn	linkedin.com/in/jpglmrodrigues
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Personal	joaor.eu